Pawn Endgames Without Edges: Exact Combinatorial Game Values, Real-Chess Certification, and Tablebases on the Cylindrical Chessboard

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Abstract

Elkies showed that king-and-pawn endgames realise a rich zoo of combinatorial game values once a passed pawn is scored as a decisive $\pm\mathsf{B}$ and a mutual-zugzwang king group is treated as 0. We study this programme on the *cylindrical* chessboard, where files a and h are glued and the board has no edges, and we ask which of its conclusions survive real chess.

(i) **The model, made exact.** We show that the textbook passed-pawn test is unsound under en passant and adopt an *e.p.-aware* test. We prove a cut lemma (a cylinder with an empty file is a flat strip), so every cylinder-only value comes from a *ring*. We give a hereditary condition under which the $\pm\mathsf{B}$ sum model predicts the true winner, and a counterexample to the condition used before.

(ii) **The $*2$ question.** Elkies asked whether $*2$ occurs in a chess position with only kings and pawns. In an exhaustive census of 231,071 flat positions (widths ≤ 4, at most 6 pawns), the only $*2$ is an artefact of the textbook test. Under the e.p.-aware test no $*k$ with $k \geq 2$ occurs. We introduce the *real-chess oracle value*, which replaces every model terminal by its tablebase outcome and bounds the result by monotonicity. With it we show that the candidate's real value is never $*2$, and we refute a second candidate outright.

(iii) **Elkies' small values.** The value $1/8$ occurs robustly: it is the same under all nine race and en passant conventions, in a reachable 10-man king-and-pawn position. So do $\uparrow^2{+}*$ and tiny(1). For the $1/8$ position, all 306 tablebase-resolvable end positions are now settled.

(iv) **Kings.** We state conditions under which "pawn components plus a frozen trébuchet" equals chess. The paper's own king group 7.2 is not a mutual zugzwang in real chess (confirmed by the Lichess tablebase). Spare king tempo moves give loopy values (on, off, dud).

(v) **Mating without corners.** We give independently replicated 4-man tablebases on the cylinder and the torus. On the cylinder king and rook cannot force mate, king, bishop and knight cannot either, and two bishops on opposite colours always can, contradicting a published claim. We also solve king and pawn against king there: 16,538 flat draws become wins and 15,930 flat wins become draws.

(vi) **Dawson's chess and engines.** We prove that a full $3 \times n$ Dawson cylinder is a first-player win iff $g(n-3) = 0$, and we correct the earlier treatment. We also report a pilot of Stockfish and Lc0 on CGT-labelled positions, and the ChessSums benchmark.

Every computational claim carries an evidence label. The code and data will be released with the paper.

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1 Introduction

Combinatorial game theory (CGT) assigns to a two-player game of perfect information a *value* that predicts the winner of any sum of such games [1, 3, 15]. Chess is not a normal-play game: the game can be drawn, and kings interact across the whole board. Even so, Elkies [6] showed that pawn endgames are approximately sums of independent local fights, and that these fights realise numbers, nimbers, infinitesimals such as $\uparrow$ and tiny(1), and hot switches. The approximation rests on two conventions:

- a side that obtains a passed pawn wins outright, which is valued as a large number $\pm\mathsf{B}$;
- a mutual zugzwang of the kings, the *trébuchet*, contributes 0.

Elkies [7] extended the zoo to higher nimbers on large boards and asked which values occur on the ordinary 8×8 board. Drummond-Cole [5] found a $*2$ on 8×8, but it needs officers (bishops, knights, rooks, queens) to immobilise the pawns. Whether $*2$ occurs with *kings and pawns only* remains open.

The cylindrical board. In *cylinder chess* files a and h are adjacent [13]. The board has no edges and every file is equivalent. Our earlier study [2] computed Elkies-style values on the cylinder. This paper revisits that work with exact software, and pushes it in four directions where, as far as we could determine, nothing has been published:

- CGT values of pawn endgames on cyclic boards;
- real-chess certification of Elkies-style values;
- exact validity conditions for the frozen-king convention;
- endgame tablebases on the cylinder and the torus.

Contributions.

1. **A precise model** (Section 3).
   - The textbook passed-pawn test lets a pawn that has just double-stepped past an enemy pawn count as passed before the legal en passant capture. We replace it with an e.p.-aware test.
   - Proved: decomposition into components, the cut lemma for cylinders, and a hereditary sum-safety theorem.
   - A counterexample to the sum-safety test used before.
2. **A corrected audit** of the 2025 paper (Section 4), checked three ways.
3. **Value atlases.** 1.2 million cylinder and flat positions and 231,071 flat 8-rank positions, with the values that occur only on the cylinder (Section 5).
4. **The $*2$ question** (Section 7).
   - The census result described in the abstract.
   - The real-chess oracle value, a method that certifies or refutes a model value with tablebases and monotonicity.
   - Its application to every $*2$ candidate.

5. **Robust small values.** $1/8$, $\uparrow^2{+}*$ and tiny(1) in reachable positions, with a partial real-chess certification of $1/8$ (Section 6).
6. **Explicit kings** (Section 8). When the frozen trébuchet is exact, where it fails, and the loopy values of spare king moves.
7. **Cylinder and torus tablebases** (Section 9). Two independent solvers agree on about 607 million values.
8. **Dawson's chess** on the cylinder (Section 10). The ring rule, and a reproduction of the misère quotient.
9. **Engines and learning** (Section 11). Stockfish and Lc0 on CGT-labelled positions, and the ChessSums benchmark.

Evidence labels. Much of this work is computational, so every claim carries one of four labels (Section 12):

- PROVED: a written proof;
- CHECKED-2: two independent implementations agree;
- COMPUTED-1: one implementation;
- CONJECTURE.

Negative results are reported as plainly as positive ones.

2 Background

2.1 Games and values

A game $G = \{\, G^L \mid G^R \,\}$ lists the positions Left (White) and Right (Black) can move to. Under *normal play* a player who cannot move loses. The disjunctive sum $G + H$ lets the player to move choose one component. Every short game has a unique *canonical form*, obtained by deleting dominated options and bypassing reversible ones, and $G = H$ iff $G - H$ is a second-player win. The outcome classes are:

- $\mathcal{L}$: Left wins whoever starts;
- $\mathcal{R}$: Right wins whoever starts;
- $\mathcal{N}$: the first player wins;
- $\mathcal{P}$: the second player wins.

We use the following values [1, 15]:

- **numbers**: the integers, and dyadic rationals such as $1/2 = \{\, 0 \mid 1 \,\}$;
- **nimbers** $*k$: $* = \{\, 0 \mid 0 \,\}$ and $*2 = \{\, 0, * \mid 0, * \,\}$;
- **ups**: $\uparrow = \{\, 0 \mid * \,\}$ and $\Uparrow = \uparrow + \uparrow$;
- **up-second** $\uparrow^2 = \{\, 0 \mid \downarrow * \,\}$, which is positive yet so small that $n \cdot \uparrow^2 < \uparrow$ for every n;
- **tinies**: $\mathrm{tiny}(x) = \{\, 0 \parallel 0 \mid -x \,\}$;
- **switches** $\pm x = \{\, x \mid -x \,\}$.

A game has a *mean* $m(G)$ and a *temperature* $t(G)$, both read from its thermograph. Positive infinitesimals such as $\uparrow$, $\uparrow^2$ and tiny(1) have mean and temperature 0, but they still decide sums.

2.2 Elkies' conventions

Convention 2.1. A position in which exactly one side has a passed pawn is worth $\pm\mathsf{B}$, with $\mathsf{B} = 256$; it exceeds every tempo count that occurs. If a move gives *both* sides a passed pawn, the result is decided by a *race rule*:

- *count*: the side that needs fewer moves to promote wins, with ties to the side that moves next;
- *mover*: the side that made the move wins;
- *opponent*: the other side wins.

Pawns change file only by capturing, so an empty file stays empty forever. The pawns between two empty files therefore form an independent *component*, and a position is approximately the sum of its components. A king group in mutual zugzwang, such as the *trébuchet*, is taken to contribute 0: whoever is forced to move a king there loses a pawn and the game. Elkies used these conventions to realise $\uparrow$, $\Uparrow *$, tiny(1), ± 1 and many fractions on ordinary boards.

2.3 The cylinder

On the cylinder $C_n \times 8$ the files are $\mathbb{Z}/n$. Kings, pawn captures, knights and sliders cross the seam between files h and a. A slider may not use a full loop of a rank as a null move (Section 9). En passant works across the seam. We write F_n for the flat board with n files.

3 The model, made precise

All values in this paper are computed by `cylcgt`. It is an exact canonical-form CGT engine with a pawn solver for flat and cylindrical boards of any width. It handles double steps, wrap-around captures, en passant, decomposition into components and the three race rules. Section 12 describes how it was validated.

3.1 En passant and the passed-pawn test

Elkies' convention needs a test for "passed". The *textbook* test calls a pawn passed if no enemy pawn stands ahead of it on its own file or an adjacent file. This is unsound when en passant is legal.

Example 3.1. White a2, Black b4. After 1.a4 the a-pawn has no enemy pawn ahead of it, so the textbook test calls it passed and scores the position $+\mathsf{B}$. But Black may reply 1…b4×a3 e.p.

Definition 3.2 (e.p.-aware passed test)**.** A pawn is *passed* if no enemy pawn stands ahead of it on its own file or an adjacent file, *and* it cannot be captured en passant on the opponent's next move. The pawn may be captured if it has just made a double step, the square it skipped is the e.p. target, and an enemy pawn stands beside it.

From Section 5 on, all results use Definition 3.2 unless they are marked *textbook*. This choice changes 21,441 of the 231,071 values in our flat census (Section 7), 4,068 of them in outcome. It is the change that removes the $*2$ candidate.

3.2 Components and the cut lemma

Lemma 3.3 (components; PROVED)**.** *An empty file stays empty. On F_n the components are intervals of occupied files. On C_n they are arcs, or one* ring *occupying every file. An arc component is isomorphic, as a game, to the same pawns on a flat strip of its own width.*

Theorem 3.4 (cut lemma; PROVED, CHECKED-2)**.** *A position on C_n with an empty file has the same value as the flat position on F_{n-1} obtained by cutting the cylinder at that file. Hence every value that occurs on C_n but not on F_n is the value of a ring position, and a ring on C_n needs at least n pawns.*

Proof sketch. The empty file can never be entered, since pawns change file only by capturing a pawn already on the target file. So no move ever crosses it. Cutting there changes no legal move and no passed-pawn test. □

The cut lemma explains why, with at most five pawns, cylinders of width 5 to 8 add no values beyond the flat strips (Section 5).

3.3 When is the $\pm\mathsf{B}$ sum model right?

In chess the first passed pawn ends the game. In a CGT sum, $+\mathsf{B}$ in one component and $-\mathsf{B}$ in another cancel. Write $S(X)$ for the sum model of a position X, and $W(X)$ for the whole-board game, which ends at the first passed pawn anywhere.

Definition 3.5. X is *hereditarily sum-safe* (HSS) if no non-terminal position reachable from X has two different components carrying passed-pawn threats of opposite sign.

Theorem 3.6 (PROVED; tested on 1,500 random positions)**.** *Suppose X is HSS, no position with two or more components carries a usable en passant right (condition (E)), and* B *exceeds the number of moves either side can make. Then $S(X)$ and $W(X)$ have the same outcome.*

Proposition 3.7 (CHECKED-2)**.** *The earlier sum-safety test inspects only the top-level components, and it is not sufficient. On C_8 with en passant off,* `W b2 d4 g3 h5 / B a7 c7 f7 g4` *passes that test, yet S predicts $\mathcal{R}$ while W is $\mathcal{N}$. The failure comes from cancellation inside sums that appear later in the game tree.*

Proposition 3.8 (COMPUTED-1)**.** *Without (E) the sum can be wrong, even in outcome. On the flat 6×6 board,* `W a4 f4 / B b4 e5` *with Black's e.p. right on a3 is the whole-board game $\{\, 0 \mid -4,\ -3 \pm 2 \,\}$ ($\mathcal{N}$), while the sum of its components is $\{\, -1* \mid -5* \,\}$ ($\mathcal{R}$).*

4 The 2025 paper, re-audited

Every pawn-only claim of Bhattacharjee [2] was recomputed. Each diagram was re-transcribed blind and compared with a pixel scan. Each verdict below was then checked by three independent reviewers. Table 1 shows the corrected verdicts. None of them changes when the textbook passed test is replaced by the e.p.-aware test.

What stands is the method: decomposition, canonical values, and the use of numbers, infinitesimals and temperature. The f-file, c-file, tiny(1), $*$ and trébuchet computations are right.

Pos.	Claimed	Computed	Verdict
2.1 e4/e5	0	0	correct
2.2 a2,h1/a3,h4	0	0	correct (the pawn on h1 is illegal in chess)
3.1 a3,h1/a4,h5	1	1 with e.p., 2 without	correct only through wrap-around en passant
3.2 a4,h1/a5,h6	2	2	value correct; the stated option list is not
4.1 e4/e6	$*$	$*$	correct
4.2 e3/e6	$\uparrow$	0	the diagram is one rank off; e2/e5 gives $\uparrow$
4.3	tiny(1)	tiny(1)	correct
5.1 trébuchet	0	mutual zugzwang	correct, only because of the seam; real in chess (Section 8)
7.1 c-file	± 1	± 1	correct
7.1 f-file	$\{\{2 \mid 1\} \mid 0\}$	same	correct (mean 3/4)
7.1 a/h-files	1	$\{+\mathsf{B} \mid *\}$	wrong: the position is hot
7.2 pawns	± 1	infinitesimal, mean 0	wrong
7.2 kings	0	zugzwang in the model only	**not** a mutual zugzwang in chess (Section 8)
7.3	2	decided at move 0	wrong: a5 and f7 are already passed
7.3 {c,d}	1	tiny(1)	wrong
§1 rules	knight h1→a1, a2	a3, b2	wrong: a1 and a2 are extra *king* squares
§8 Dawson	recurrence	see Section 10	wrong: no first move exists under the stated rule

Table 1: Audit of the pawn-only claims of the 2025 paper (CHECKED-2).

5 Value atlases: the flat board and the cylinder

5.1 Scope

- **Cylinder atlas.** Every canonical pawn position with $w + b \leq 5$ pawns on C_n and F_n, $n = 3, \ldots, 8$, with 8 ranks: 1,223,689 positions and 10,369 distinct values (textbook test).
- **Flat census.** The 8-rank flat census used for the $*2$ search (Section 7) covers widths 1 to 4 and $w + b \leq 6$: 231,071 records, standing for 459,443 board positions.

5.2 What the cylinder adds

By Theorem 3.4, cylinder-only values come from rings, so with at most five pawns they live on C_3, C_4 and C_5.

- **Cylinder-only values** (COMPUTED-1; outcomes CHECKED-2 by an independent minimax). 53 values free of B occur on some cylinder of width ≤ 8 but on no flat strip of width ≤ 8:
  - 40 come from 3-file rings, for example 3/4 at `W a2 b4 / B b7 c6` and $1/4 \pm 1/4$ at `W a2 b2 / B c5 c6`;
  - 13 come from 4-file rings, for example $\{3/2 \pm 1/2 \mid -3/2 \pm 1/2\}$ at `W a2 c2 / B b7 d7`.
- **The converse is empty.** No value occurs on a flat strip and on no cylinder.

Remark 5.1 (a correction found by review). The atlas first claimed that no cylinder-only value is robust to the race rule. An independent re-computation refuted this:

- the 3-file ring values do depend on the race rule (3/4 becomes 0 under *mover*);
- the 4-file ring values do not, and change only when en passant is switched off.

No value is cylinder-only under all six race/e.p. settings.

5.3 Value classes

With at most five pawns (textbook test, count rule):

- **Nimbers:** $*$ and $*2$ only.
- **Ups:** $\uparrow$, $\uparrow*$, $\Uparrow$ and $\Uparrow*$ occur as single components.
- **Fractions:** $\pm 1/2$, $\pm 1/4$, $\pm 3/2$ and $\pm 5/2$ on both boards; $\pm 3/4$ only on 3-file rings.
- **Temperature:** without `B` the highest temperature is 3, from `W a2 b4 / B a5 b7` $= \pm 3$, on every width of both boards.

These classes include the paper's own examples. The six-pawn census adds 1/8, 1/16, 1/32, $\uparrow^2$ and $\uparrow^2*$ (Section 6).

6 Elkies' small values

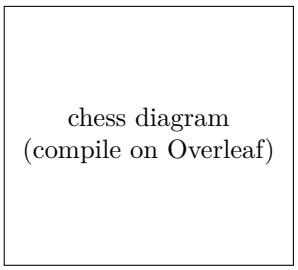
Elkies realised $\uparrow$, tiny(1) and many fractions, and asked which smaller values occur on an 8×8 board. Table 2 counts exact occurrences in the flat census under both passed-pawn tests. An independent C solver, which shares no code with `cylcgt`, confirms every listed identity by testing $G - T = 0$ (CHECKED-2).

Target	textbook	e.p.-aware	robust example (all nine settings)
1/8	13	9	`W a2 a3 b2 / B a6 b5 b6` and its mirror `W a2 b2 b3 / B a5 a6 b6`
1/16	3	2	—
1/32	1	0	—
$\uparrow^2 = \{\, 0 \mid \downarrow* \,\}$	5	2	—
$\uparrow^2* = \{\, 0, * \mid \downarrow \,\}$	10	10	`W a2 b2 b6 / B a6 b7 c5`
tiny(1)	13	16	`W a2 b2 b6 / B a6 b7 c6` (e.p. on)
tiny(x), $x \notin \mathbb{Z}$	0	0	—

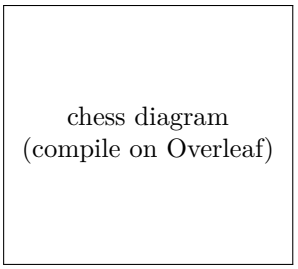
Table 2: Elkies targets in the flat census (widths ≤ 4, ≤ 6 pawns, count rule, e.p. on). "Nine settings" means the three race rules $\times$ {e.p. off, textbook, e.p.-aware}.

Theorem 6.1 (CHECKED-2)**.** *In Elkies' model,* `W a2 b2 b3 / B a5 a6 b6` *has value exactly* 1/8. *This holds under every race rule, with en passant off, textbook or e.p.-aware, and for* every *assignment of outcomes to its race positions.*

The mirror image `W a2 a3 b2 / B a6 b5 b6` has an unreachable pawn structure, but this one is reachable in a legal game: c2×b3, a7–a5, b7×a6, c7×b6. Embedded beside the trébuchet it gives the 10-man position in Figure 1.



1/8 (model): `8/8/pp6/p4Kp1/6Pk/1P6/PP6/8`



$\uparrow^2 *$ (model): `8/1p6/pP6/2p2Kp1/6Pk/8/PP6/8`

Figure 1: Robust small values beside the trébuchet (W Pg4 / B Pg5, WK f5, BK h4).

Two textbook results do not survive the e.p.-aware test.

- The earlier "best reachable 1/8", `W a2 b2 d2 / B a5 c4 d7`, becomes $-3/2 \pm 3/2$.
- The best $\uparrow^2$, `W a3 c5 d4 / B a7 b7 d7`, becomes $\{\, 3/4 \mid \downarrow * \,\}$.

In both, the value relied on a double step that the textbook test called passed.

6.1 Towards a real-chess 1/8

We computed the real-chess oracle value (Definition 7.3) of the 1/8 position in Figure 1 (CHECKED-2 at every resolved end position: tablebase or tablebase-backed search; the values are COMPUTED-1).

- **End positions with at most 7 men.** All 306 sides were resolved against the tablebase. 283 agree with the model, 5 disagree, and 2 are draws.
- **The decisive set.** A greedy minimal-support search over the open end positions showed that only 8 of them matter: 5 with 8 men and 3 with 9. Resolving their 8-man sides by tablebase-backed search gave 7 agreements and one disagreement. With Black to move in `W a3 a5 / B b4 b5`, Black wins, although the model scores a White win.
- **Bounds now.** With only exact results fixed, $-1/2 \le O \le 3/8 \pm 1/8$. If the 4 end positions still open (three with 9 men, one with 8) have the outcome the model predicts, then $O = 1/8$ *exactly*, whatever the other open end positions do. These four are being solved exactly, with every king placement, by the king-and-pawn solver of Section 8.
- **Exact results so far.** In `W b2 b5 / B a5 b4` with White to move, White wins, as the model says. With Black to move the solver, allowing one promotion, gives only the bounds draw-or-worse to win, so a run allowing two promotions is under way. The three 9-man positions are still being solved. Each of the 7 open sides is decisive on its own: if any one of them deviates from the model, the lower bound falls to 0, $\uparrow *$ or below. These runs allow queen promotions only, so a full certificate needs a final run with all four promotions.

So the value 1/8 has survived every real-chess test so far, including a disagreement at one of its decisive end positions. Resolving the last four is in progress. Success would give a king-and-pawn 8×8 position of real oracle value 1/8, conditional only on the king freeze. Elkies' constructions did not reach that value.

7 The $*2$ question

Question 7.1 (6, 7)**.** Is there an 8×8 chess position with only kings and pawns whose value is $*2$?

7.1 A candidate, and why it fails

In the flat census (widths ≤ 4, ≤ 6 pawns, count rule, e.p. on, textbook test), exactly one component has value $*k$ with $k \geq 2$: the component `W a2 c2 / B b7 d7` (Figure 2).

- **Robustness (textbook test).** It is $*2$ under the count and mover rules, with en passant on or off, and 0 under the opponent rule. Only 4 of its race positions change the value when flipped alone.
- **Embedding.** The game tree never splits into components before a passed pawn appears, so the sum model equals the whole-board game (PROVED). The embedding with the g-file trébuchet has no checks and no stalemates, and every king move drops that side's g-pawn (PROVED; checked exhaustively).
- **The e.p.-aware test.** The value becomes $\{\, 2 \pm 1 \mid -2 \pm 1 \,\}$ (CHECKED-2). The reason is that after a2–a4 Black has b4×a3 e.p.

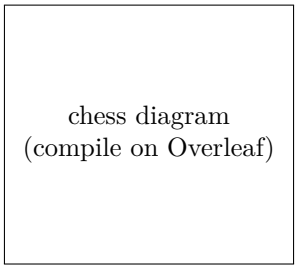


Figure 2: The $*2$ candidate: `8/1p1p4/8/5Kp1/6Pk/8/P1P5/8`.

Theorem 7.2 (census; CHECKED-2)**.** *Under the e.p.-aware test (count rule, e.p. on), no position in the flat census has value $*k$ with $k \geq 2$. The nimber column has* 6,724 *zeros,* 4,042 *stars and no* $*2$.

Two independent programs agree on the outcome and nimber of all 231,071 records under both tests: the CGT engine and a C solver that tests whether $G + *k$ is a second-player win. No $*3$ or higher occurs under any of the nine settings. Every $*2$ that occurs under some setting has 4 or 6 pawns, so none fits in a 7-man tablebase together with the trébuchet.

7.2 The real-chess oracle value

A model value is a statement about a convention. To test it against chess we keep the pawn-move tree of the model and replace only its leaves.

Definition 7.3 (real-chess oracle value)**.** Let X be a component, embedded in an 8×8 position beside a king freeze.

- Every *terminal* of the model tree, the first position in which some side has a passed pawn, is replaced by a game built from its real results r_W, r_B (White moving first, Black moving first):
  - $\mathcal{L} \mapsto +\mathsf{B}$ and $\mathcal{R} \mapsto -\mathsf{B}$;
  - $\mathcal{N}$, $\mathcal{P}$ and draws are given a *treatment*: a game in $[-2\mathsf{B}, 2\mathsf{B}]$.
- The canonical value of the resulting tree is the *oracle value* $O(X)$.

Lemma 7.4 (sandwich; PROVED)**.** *Raising an option never lowers a game. So if the unresolved terminals are set to* $-2\mathsf{B}$ *and* $+2\mathsf{B}$*, the two resulting values bound* $O(X)$ *for every treatment and every unknown outcome. When the bounds coincide,* $O(X)$ *is certified.*

The oracle value separates two questions: is the pawn convention right at the leaves, and is the king freeze right? The second is the subject of Section 8.

Theorem 7.5 (CHECKED-2 at the leaves; values COMPUTED-1)**.** *For the candidate embedded as in Figure 2:*

(a) In real chess the root is a win for the side to move, with either side to move ($\mathcal{N}$*, proved by explicit certificates). With Black to move only 1... b5 wins. With White to move 1.c4 wins and every king move loses.*

(b) In the e.p.-aware tree, $O(X) \neq *2$ *and* $O(X) \neq 0$ *under* every *treatment: the bounds force Left's stop to be at least* 1 *and Right's at most* -1*.*

(c) If draws count as wins for Black, $O(X) = \{\, 2 \pm 1 \mid -2 \pm 1 \,\}$ *exactly, in both trees.*

So the textbook $*2$ does not survive real chess. Of the other candidates, each a $*2$ under one non-default rule and each needing 10 men:

- `W a3 b3 c2 / B b5 b7 c6` is refuted: its oracle value is positive under every treatment (CHECKED-2);
- for three others the tablebase contradicts the race verdicts that their $*2$ needs, but their oracle values cannot be pinned with 7-man tablebases.

Status of the question. Within flat components of width ≤ 4 with at most six pawns, no $*2$ survives either the adopted convention or real chess. The question remains open beyond that scope; Section 13 lists the next searches.

8 Explicit kings: when is the frozen trébuchet exact?

The convention treats a king group as 0; Mäser [10] already warned that this is heuristic, and Siegel [14] studied loopy chess positions. We put the kings back with an exact king-and-pawn solver, `kpsolve`. It plays real chess with kings, pawns, en passant, all four promotions and stalemate, and no fifty-move rule. An independent reference solver agrees with it on all 1,418,480 positions of 18 small-board runs. On every comparable position it is consistent with the Lichess tablebase (CHECKED-2).

8.1 Validity conditions

Theorem 8.1 (PROVED; reviewed)**.** *Let a position consist of pawn components and a king group. Its chess outcome equals the normal-play outcome of the pawn components if all of the following hold:*

(V1) the kings never interact with the pawn components, and nobody is ever stalemated;

(V2) every decisive event *(the appearance of a passed pawn) really wins;*

(V3) every move inside the king group is decisively punished.

The step from there to the $\pm\mathsf{B}$ *arithmetic needs one more condition:*

(V4) no two components carry passed-pawn threats of opposite sign, and B *exceeds every tempo count.*

(V3) has a local certificate: the punishing side promotes within T moves even if the defender may pass, while the defender cannot promote within T moves.

- With $T = 1$ the certificate holds in every context (PROVED).
- For $T \geq 2$ we conjecture that it fails in some context (CONJECTURE). The counterexample first offered turned out, on independent review, to be a terminal position, so it does not settle the question.

Real trébuchets have $T \approx 5$–7. So the certificate alone does not show that their "zero" is valid in every context; we expect it to hold only beside slow or locked components.

8.2 Spare tempo moves are loopy

A king that can shuttle forever without loss is not 0 but a loopy game:

- $\mathsf{on} = \{\, \mathsf{on} \mid \}$ if only White can shuttle;
- $\mathsf{off} = \{ \mid \mathsf{off} \,\}$ if only Black can;
- dud if both can.

Since on exceeds every number, B included, the B arithmetic breaks down completely there.

Theorem 8.2 (PROVED)**.**
- *If only White can pass, Black wins iff Black can force a decisive event against a passing White; otherwise White wins.*
- *If both sides can pass, a side wins iff it can force a decisive event against a passing opponent; otherwise the game is drawn.*

8.3 The paper's king groups in real chess

Group	Board	White to move	Black to move	Verdict
5.1: Kh5 Pa4 / Kb4 Pa5	cylinder	Black wins	White wins	real mutual zugzwang (CHECKED-2)
Figure 2: Kf5 Pg4 / Kh4 Pg5	flat	Black wins	White wins	real mutual zugzwang (Lichess)
7.2: Ke6 Pd7 / Kd8 Pe7	flat	**draw**	White wins	**not** a zugzwang (Lichess)

Table 3: King groups standing alone. The convention reads each as 0.

In 7.2, White to move draws with Kd5, Ke5 or Kf5: the d7 pawn falls, but the white king catches the e-pawn. Only Kf7 loses. The audit's statement that 7.2 is a mutual zugzwang is therefore true only inside the convention.

8.4 How often does the convention fail?

In sweeps over every small pawn component beside a trébuchet, on 6×6 and 7×6 cylinders and a 5×6 flat board, 159 of 500 testable components get the wrong outcome (COMPUTED-1).

- In 107 the kings alone are the cause: with frozen kings the solver reproduces `cylcgt` exactly.

- The other 52, all on 7×6, are failures of the pawn convention itself: 18 through en passant, and 34 because a passed pawn does not actually win its race.

Small boards are a stress test, since the kings are never far from the pawns. On 8×8 the failure rate is lower but not zero (Section 7).

9 Mating without corners: tablebases on the cylinder and the torus

Without corners or edges, can the standard material still force mate? We computed all 4-man pawnless tablebases on the plane, the cylinder (files a and h glued) and the torus (also ranks 1 and 8 glued), using the Lichess tablebase only for comparison on the plane [9].

- **Rules.** Sliders stop when their ray returns to their own square, so there is no null move round a rank. There is no castling and no fifty-move rule. Stalemate is a draw.
- **Verification.** Two independently written retrograde solvers agree on every position of all 24 tables, about 607 million values with either side to move. One of them also checks the Bellman equations and the board symmetries on every position (CHECKED-2). On the plane both reproduce the known maxima: KQK 10, KRK 16, KBBK 19, KBNK 33.

	plane		cylinder		torus	
Material	won	max	won	max	won	max
KQK	100%	10	100%	10	0	—
KRK	100%	16	**2.80**%	3	0	—
KBBK (opposite colours)	99.98%	19	**100**%	**14**	0	—
KBNK	99.51%	33	**0.0245**%	2	0	—
KNNK	0.0107%	1	0.0077%	1	0	—
KRBK	100%	16	99.74%	13	0.0218%	1
KRRK	100%	7	100%	8	100%	7
KQRK	100%	6	100%	5	100%	5

Table 4: Share of legal positions won with White to move, and the longest forced mate in moves.

Findings.

- **KRK and KBNK are draws on the cylinder.** Without an edge to drive the king against, the rook and the bishop-knight pair cannot force mate. The few wins are positions that are already almost mates.
- **Two bishops on opposite colours win every position on the cylinder**, in at most 14 moves. This contradicts a published claim by H. G. Muller that the bishop pair cannot force mate there [11].
- **On the torus no single officer can mate, nor any pair below two rooks.** Some mates of the bishop pair exist on the torus but can never arise in play. Only KRRK and KQRK win.

We found no earlier tablebase for these boards.

9.1 King and pawn against king

We solved every legal KPK position on the 8×8 plane and cylinder twice: with the king-and-pawn solver of Section 8 (all four promotions) and with a separately written KPK solver (queen promotion only). On the cylinder the two agree on all 326,640 positions. On the plane they differ on exactly 6 positions, the classical ones where only a rook promotion avoids stalemate (CHECKED-2).

	White to move		Black to move	
Board	legal	White wins	legal	White wins
plane	163,328	124,960 (76.51%)	168,024	97,604 (58.09%)
cylinder	160,656	126,736 (78.89%)	165,984	92,760 (55.88%)

Table 5: KPK on the 8×8 plane and cylinder. The cylinder has fewer legal positions because the kings may not stand next to each other across the seam.

- **The seam cuts both ways.** Of the 326,640 positions legal on both boards, 16,538 flat draws are cylinder wins: the rook-pawn fortress in the corner no longer exists, as in Ka6, Pa5 against Ka8. In the other direction, 15,930 flat wins are cylinder draws.
- **More mutual zugzwangs.** The cylinder has 216 KPK positions in which White to move only draws but Black to move loses, against 160 on the plane.

10 Dawson's chess on the cylinder

Dawson's chess [4] is played on a $3 \times n$ board:

- White pawns fill rank 1, Black pawns fill rank 3, and rank 2 is empty;
- pawns move as in chess, and captures are compulsory;
- a player who cannot move loses (normal play).

Guy and Smith [8] identified normal play with the octal game 0.137. Its Grundy values $g(n)$ have period 34, with last exception $n = 51$ (classical; reproduced for $n \leq 300$, CHECKED-2).

Correction. The 2025 paper's §8 is wrong in three ways (CHECKED-2):

- its move rule allows no first move;
- its recurrence omits the end-file move and $g(1) = 1$, and agrees with the true g at only 31 of the values $n = 0, \ldots, 300$;
- its cylinder predictions are therefore wrong for 56 values of $n \leq 300$.

Theorem 10.1 (ring rule; PROVED, machine-checked for $n \leq 22$)**.** *On the full $3 \times n$ cylinder with $n \geq 3$, every first move forces exactly five plies of captures on the pushed file f. All lines end in the same position: files $f \pm 1$ empty, a blocked pair on f, and a full flat strip of $n - 3$ files with the opponent to move. Hence the ring's Grundy value is* $\operatorname{mex}\{g(n-3)\}$*, and the first player wins iff $g(n-3) = 0$, that is for $n = 3, 7, 11, 17, 23, 27, 31, 37, 41, 45, 57, \ldots$*

Misère play. Dawson posed the game in misère form, where the last player to move loses. That version is still unsolved.

- An independent implementation of the Plambeck–Siegel quotient algorithm [12] reproduces their partial quotient for strips of length ≤ 32 exactly: order 638, with 109 P-positions. It agrees with exhaustive search on 7.4 million positions (CHECKED-2).
- The next quotient did not converge within our budget.
- The partizan values that a direct solver produces for each board are not additive (for example $D_5 = *$ while $g(5) = 3$). The additive invariant is the impartial macro-move value.

11 Engines and learning

A CGT test suite. We built 233 legal 8×8 positions with at most 7 men, each of a known CGT class. All were checked against the Lichess tablebase: in 181 the CGT-predicted winning moves are confirmed, and in the other 52 CGT and the tablebase disagree.

Stockfish 19 (official build, 1 thread, no tablebases).

- It solves all 233 positions at 10^5 nodes, and 232 already at 10^3.
- At shallow fixed depth, accuracy depends on the CGT class, as CGT predicts (Table 6). Infinitesimals and nimbers are tempo and parity fights that the static evaluation misreads; hot positions and numbers are read correctly early.

Lc0 (bundled network, Metal backend).

- It is perfect at 10^3, 10^4 and 10^5 nodes in every class, including all 17 draws.
- The suite is therefore too easy to separate the engines. A real contrast needs sums beyond 7 men, where only CGT supplies the ground truth.

CGT class	n	depth 1	2	4	6	8	12
hot (±B switch)	68	81%	78%	91%	97%	99%	100%
number	57	79%	81%	84%	93%	95%	98%
switch	37	73%	78%	89%	100%	100%	100%
infinitesimal	27	**30%**	**26%**	**59%**	85%	100%	100%
nimber	17	**53%**	65%	88%	100%	100%	100%

Table 6: Stockfish 19: share of evaluations that match the tablebase result, by CGT class and search depth.

ChessSums. A benchmark of pawn-endgame sums with exact ground truth, for evaluating engines and language models.

- **Components.** The library has 418 components with 37 distinct values, each with the same value under both passed-pawn tests, all race rules and en passant on or off.
- **Data.** 26,000 rows. Every outcome was confirmed by whole-board alpha–beta, and 4,887 also by an independent minimax.

- **Splits.** Training uses at most three components and testing four to six; some component types are held out.
- **Baseline cost.** Whole-board alpha–beta that ignores the decomposition grows by a factor of about 4–9 in nodes per added component, while the CGT sum stays instant.
- **Reward.** A draft reward function for reinforcement-learning fine-tuning is included.

12 Verification and reproducibility

Evidence labels.

- PROVED: a written proof. Proofs were drafted with AI assistance. Each was then re-derived independently by a second, different model: 8 of 9 were confirmed, one after a repair, and one gap was found and is reported as a conjecture. A human check is still needed.
- CHECKED-2: two independently written programs agree. "Independent" means written from the rules alone, without reading the other's code.
- COMPUTED-1: a single implementation.
- CONJECTURE: neither proved nor computed.

Cross-checks of the CGT engine.

- A brute-force sum solver agrees with `cylcgt` on 4,967,550 outcome predictions, and on 2,483,775 more under the e.p.-aware test. These cover 55,195 positions on 3–5 files, three race rules, and fifteen reference games X (the prediction is outcome$(G + X)$).
- A whole-board minimax agrees on all 1,070 single-component positions tested.
- The independent C solver of Section 7 agrees on all 231,071 census records.
- A third implementation was written by a different AI system (OpenAI Codex) from the rules alone, in plain Python, without importing any of the project's code. It reproduces the headline values of Sections 5 and 7 in all nine settings (three race rules, e.p. off, textbook and e.p.-aware). It finds no $*k$ with $2 \le k \le 292$ among 793,299 labelled placements of up to six pawns on 1–4 files; 292 bounds the birthday of every such game (36 pawn moves, then a terminal of birthday 256).

Real chess.

- **Tablebase.** All positions with at most 7 men were probed in the Lichess tablebase (Syzygy), with every response cached.
- **Move generator.** Our legal-move generator matches Lichess's move lists on 160 of 160 test positions, covering en passant, promotions, pins, checks, mates and stalemates.
- **Larger positions.** Positions with 8 or more men were resolved by exhaustive search with tablebase look-ups at every capture.

Use of AI tools. The software, the searches and the first drafts of several proofs were produced with AI coding assistants (Claude, Codex and others) under the author's direction. Every headline number has an exact reproduce command in the accompanying repository, and the label of each claim records how strongly it was checked.

Code and data. Everything is in the project repository (to be released with the paper):

- `cylcgt/`: the CGT engine;
- `atlas/`: the value atlases;
- `realchess/`: the oracle value;
- `kings/`: explicit kings;
- `tablebase/`: the tablebases;
- `dawson/`: Dawson's chess;
- `engine/` and `bench/`: engines and ChessSums.

13 Open problems

1. $*2$ **with kings and pawns.** Extend the census to width 5 and to 7–8 pawns, and to rings on wider cylinders, under the e.p.-aware test. A $*2$ at most 3 files wide could be frozen on the 8-file cylinder without touching the kings' files.
2. **A certified** $1/8$**.** Resolve the 8 decisive end positions of Section 6.1. A positive answer gives a king-and-pawn 8×8 position of real oracle value $1/8$, conditional only on the king freeze. The freeze is then checkable with `kpsolve`-style search.
3. **The freeze on** 8×8**.** Turn the local certificate of Theorem 8.1 into an automatic proof that a given trébuchet is 0 beside a given component.
4. **New surfaces.** Pawn games and tablebases on the Möbius strip and the Klein bottle. Pawns crossing a twisted seam reverse direction, and no CGT of such boards exists.
5. **Odd circumference.** On cylinders with an odd number of files, king opposition parity breaks. Classify the trébuchets and the KPK results there.
6. **King-and-pawn tablebases on the cylinder.** The classic rook-pawn draw disappears there. Give a complete KPK and KPKP theory without edges.
7. **Thermography and atomic weights** of the atlas: which temperatures and atomic weights does the cylinder realise?
8. **Misère Dawson's chess.** Is the misère quotient infinite? What is the misère ring rule?
9. **Engines and language models on ChessSums** beyond 7 men, where only CGT knows the answer.

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A Positions referenced in the text

Pawn lists name the White pawns, then the Black pawns. Embedded positions add the trébuchet W Pg4 / B Pg5, WK f5, BK h4, and are given as FEN with White to move.

Role	Pawns	FEN / remark
$*2$ candidate (textbook)	`W a2 c2 / B b7 d7`	`8/1p1p4/8/5Kp1/6Pk/8/P1P5/8 w`
refuted candidate	`W a3 b3 c2 / B b5 b7 c6`	`8/1p6/2p5/1p3Kp1/6Pk/PP6/2P5/8 w`
robust 1/8	`W a2 b2 b3 / B a5 a6 b6`	`8/8/pp6/p4Kp1/6Pk/1P6/PP6/8 w`
robust $\uparrow^2 *$	`W a2 b2 b6 / B a6 b7 c5`	`8/1p6/pP6/2p2Kp1/6Pk/8/PP6/8 w`
tiny(1)	`W a2 b2 b6 / B a6 b7 c6`	e.p. on, every race rule
7.2 king group	—	`3k4/3Pp3/4K3/8/8/8/8/8`: White to move draws
5.1 trébuchet (cylinder)	—	`8/8/8/p6K/Pk6/8/8/8`: mutual zugzwang across the seam
sum-safety counterexample	`W b2 d4 g3 h5 / B a7 c7 f7 g4`	C_8, e.p. off